

PROJECT AND TEAM INFORMATION

Project Title

(Try to choose a catchy title. Max 20 words).

Intelligent Emergency Response and Traffic Network Optimization for Smart Cities

Student / Team Information

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PROPOSAL DESCRIPTION (10 pts)

Motivation (1 pt)

(Describe the problem you want to solve and why it is important. Max 300 words).

Emergency vehicles such as ambulances, fire trucks and police vehicles need to reach their destinations quickly, especially when responding to accidents, medical emergencies or other critical situations. In a city, however, traffic congestion can make this difficult. Emergency vehicles have to share the same roads with normal traffic and may also face delays at signalized intersections. Even a short delay at several consecutive signals can increase the overall response time.

Most conventional traffic signals are mainly designed to manage the regular movement of vehicles. Although some systems use adaptive signal control or emergency vehicle priority, the complete journey of an emergency vehicle is not always considered as one connected problem. Route selection, traffic conditions and signal priority may be handled separately. This can make it difficult to provide a smooth path for an emergency vehicle through a busy road network.

The motivation behind this project is to explore how algorithms and data structures can be used to address this problem through a software-based solution. The proposed system will represent the city road network as a graph, identify a suitable route for an emergency vehicle and provide priority at selected intersections along the route. The initial system will be developed in a simulated environment, allowing different traffic situations to be tested without requiring physical traffic-light hardware.

State of the Art / Current solution (1 pt)

(Describe how the problem is solved today (if it is). Max 200 words).

At present, emergency vehicles mainly use the existing road network along with conventional traffic signals. In normal situations, traffic signals operate according to fixed timings or adapt to the traffic volume at an intersection. However, these systems may not always respond quickly when an emergency vehicle is approaching.

Some existing solutions provide emergency vehicle priority by changing traffic signals to create a clear path. Route planning systems can also help emergency vehicles select a shorter or faster route based on the available road network. More advanced approaches combine traffic information, communication between vehicles and infrastructure, and adaptive signal control.

However, route planning and traffic signal management are often considered as separate functions. Our project focuses on bringing these ideas together in a single software-based model, where an efficient emergency route can be selected and traffic priority can be considered along that route.

Project Goals and Milestones (2 pts)

(Describe the project general goals. Include initial milestones as well any other milestones. Max 300 words).

The main goal of this project is to develop a software-based system that can improve the movement of emergency vehicles through a city road network. The system will focus on reducing unnecessary delays by combining route selection with traffic signal priority. Instead of treating the road network and traffic signals as separate parts, the project will study how they can work together to provide a more efficient path for an emergency vehicle.

The first milestone is to study the existing approaches used for emergency vehicle routing, traffic signal control and emergency vehicle priority. This will help in understanding the limitations of current solutions and deciding the main features of our system. The second milestone is to design the road network as a graph, where roads and intersections can be represented using suitable data structures.

The next milestone is to implement a route optimization module that can find an appropriate path between the emergency vehicle and its destination. After this, an emergency priority mechanism will be added so that selected traffic signals can be coordinated along the chosen route. The system will then be tested using different road layouts and traffic conditions.

Finally, the results will be compared using factors such as route distance, estimated travel time and waiting at intersections. The project will also leave scope for future improvements such as real-time traffic data, multiple emergency vehicles and more advanced traffic prediction methods.

Project Approach (3 pts)

(Describe how you plan to articulate and design a solution. Including platforms and technologies that you will use. Max 300 words).

The project will be developed as a software-based model of a smart city road network. The first step will be to represent roads and intersections using a graph, where intersections can be treated as nodes and roads as edges. This representation will allow the system to work with different road layouts and apply suitable route optimization algorithms.

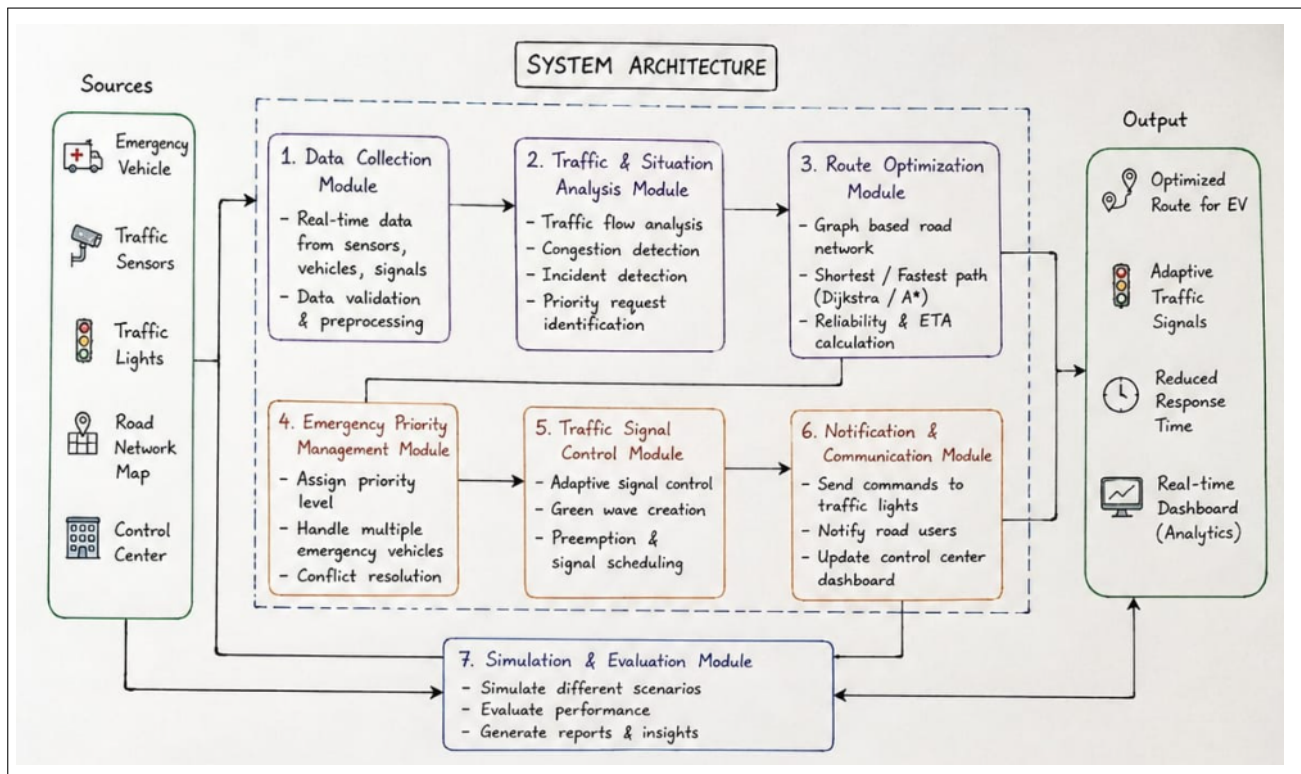
When an emergency vehicle request is received, the system will identify its starting point and destination and use the road network to find an efficient route. Traffic conditions and road-related information can be considered while selecting the route. Once a route is selected, the system will identify the important intersections through which the emergency vehicle needs to travel and apply the required traffic priority mechanism.

The project will mainly use programming, data structures, graph algorithms and simulation. The implementation can be developed using C++ or Python depending on the requirements of the later stages. A simulated road network will be used instead of physical traffic lights and roadside hardware. Different traffic conditions and emergency scenarios will be created to test the system.

The performance of the proposed approach will be studied using factors such as travel distance, estimated response time and waiting time at intersections. The results can then be used to identify limitations and possible improvements for future phases.

System Architecture (High Level Diagram)(2 pts)

(Provide an overview of the system, identifying its main components and interfaces in the form of a diagram using a tool of your choice).



Project Outcome / Deliverables (1 pts)

(Describe what are the outcomes / deliverables of the project. Max 200 words).

The main outcome of the project will be a software-based prototype that demonstrates how emergency vehicles can be guided through a city road network with reduced delays. The system will provide a way to represent roads and intersections, identify an efficient route between two locations and manage priority for an emergency vehicle at selected traffic signals.

The project will also include a simulated environment in which different road layouts, traffic conditions and emergency situations can be tested. This will make it possible to compare the normal route with the optimized route and observe the effect of traffic signal priority on response time.

The final deliverable will be a working prototype along with the project documentation, system architecture and test results. The project will also provide a base for future improvements such as real-time traffic information, multiple emergency vehicles, dynamic route changes and more advanced traffic management techniques.

Assumptions

(Describe the assumptions (if any) you are making to solve the problem. Max 100 words)

The project assumes that the road network and basic traffic information are available in digital form. Emergency vehicles are assumed to be identifiable by the system and their starting point and destination are available. Traffic signals are assumed to be capable of receiving priority instructions within the simulated environment. The initial development will focus on a software-based model and will not depend on physical traffic lights, sensors or other hardware. The traffic conditions used for testing will be simulated and may not represent every real-world situation.

References

(Provide a list of resources or references you utilised for the completion of this deliverable. You may provide links).

1. Shaaban et al., "A Strategy for Emergency Vehicle Preemption and Route Selection," 2019.
2. Dodia, A., Kumar, S., Rani, R., Pippal, S. K., and Meduri, P., "EVATL: A novel framework for emergency vehicle communication with adaptive traffic lights for smart cities," *IET Smart Cities*, vol. 5, pp. 254–268, 2023. DOI: 10.1049/smc2.12068.
3. Dijkstra, E. W., "A note on two problems in connexion with graphs," *Numerische Mathematik*, vol. 1, pp. 269–271, 1959. DOI: 10.1007/BF01386390.
4. Federal Highway Administration, "Traffic Signal Timing Manual: Chapter 9 – Advanced Signal Timing Topics," U.S. Department of Transportation.